

Caching Policies

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Learning Objectives

Be able to...

- Describe and simulate FIFO, LRU, and optimal scheduling algorithms
- Describe the tradeoffs of write-through and write-back caching and when to use them
- Describe when to use TTL policies

Recap

- What's a working set?
- What is temporal locality?
- How does caching work (see image from lecture)
- How might we estimate the working set?

FIFO, LRU, and optimal caching policies

- Suppose that you could see the future, what would you do?
 - Choose the block whose next use is farthest in the future
 - This is called Belady's optimal (OPT) algorithm
 - We use this as a comparison point: it tells us the best possible cache performance.
 - We cannot actually see the future!
- Key Idea: assume that past behavior is indicative of future behavior:
 - FIFO: evict the entry that was inserted into the cache earliest
 - LRU: evict the entry that was used least recently.
- Exercise, given the trace, simulate a cache with size 3 when using LRU, FIFO, and OF

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